

Metropolitan Business Feasibility - Executive Summary

Metros analysed: 386
Business categories: 9
CBP rows analysed: 369,496

Data stack: CBP 2023 + ACS 2024 + BLS 2024 + BEA RPP 2024

Model Performance (test set):

Best classifier: LogisticRegression (tuned)
Test Accuracy: 0.8333
Test F1: 0.7451
Test ROC-AUC: 0.9020
Best regressor: RandomForestRegressor
Test R-squared: 0.5128

Top 10 Metros (by Opportunity Score, post-RPP-adjustment):

1. San Jose-Sunnyvale-Santa Clara, CA	score=0.723
2. Midland, TX	score=0.617
3. Bozeman, MT	score=0.597
4. San Francisco-Oakland-Fremont, CA	score=0.577
5. Sioux Falls, SD-MN	score=0.571
6. Bismarck, ND	score=0.571
7. Casper, WY	score=0.570
8. Bridgeport-Stamford-Danbury, CT	score=0.570
9. Minot, ND	score=0.560
10. Rapid City, SD	score=0.556

Key insight: BEA Regional Price Parities re-shaped the rankings.

- San Francisco fell from #2 to #4 (cost-of-living index 116).
- Boston fell from top-5 to #13 (CoL 108).
- Midland TX rose to #2 (CoL 96).
- Sioux Falls and Bismarck entered top 6 (CoL ~91).

Generated automatically from outputs/* by exec_report.py